

Retrieval Augmented Generation AI POC Evaluation

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Date: Apr 8, 2026

Executive Summary:

This report presents a Retrieval-Augmented Generation (RAG) proof of concept that answers AI/ML questions for two internal audiences: several hundred engineers and a 40 person marketing team. After evaluating 13 pipeline configurations across 75 labeled question-answer pairs, I found that a minimal retrieval stack (no query rewriter, no reranker, small chunks) paired with the Cohere Command A LLM as generated the best results: BERT-F1 of 0.876/0.881 (research/marketing), faithfulness scores above 0.87 for both personas, and 3.5x faster inference than the local Mistral-7B alternative. I recommend a pay-per-use API deployment for production, with chunk size tuning as the highest-impact next step.

Introduction:

The goal of this AI POC was to determine if a RAG application of an LLM model at our organization would help both our Engineering and Marketing departments to help organize the Q&A, particularly as we help roll out new cutting edge AI Products. To accomplish this task, we evaluated 2 LLM models (an open sourced model from [Mistral](#) and a closed sourced model from [Cohere](#)) against a series of questions from the Generative AI knowledge space that teams from each department are looking to answer. The system we constructed takes this technical knowledge base covering wikipedia, published papers on arXiv and blog posts from leading researchers to produce an answer suitable for use by each department. We introduce this as a “Dual Persona” pipeline as it is able to produce suitable answers for both one of our Engineers and one of our Marketers. Each question is therefore taken into this pipeline, split into instructions specific to the needs and expertise of these two personas, and returned as answers that can be used to either help us Go To Market with a new product, or research and apply new technologies from the latest RAG findings.

Key Findings:

1. The closed source model from Cohere was about 4x faster during inference than the Open Source Model from Mistral.
2. Using rewriter and reranker steps in the query pipeline performed worse than a more simple configuration that did not include those two additional query process steps. More sophisticated optimizations seemed to degrade performance which was an unexpected result.
3. Chunk size of 128 tokens produced strong aggregate scores but shaky detail retrieval. The small chunk size won the experimental evals on average metrics, but test question analysis revealed it truncates context mid-sentence, causing the model to miss specific terms (e.g., "DPR" for question 0). Larger chunks (500-1000) would likely improve precision on detail-oriented questions.
4. BERTScore is the more robust metric for the dual-persona evaluation I set up. ROUGE systematically penalizes the marketing persona for using audience-appropriate vocabulary that

differs from gold references. BERTScore correctly captures the semantic equivalence across style shifts.

5. The model handles out-of-domain questions quite well. When asked about Marvel actors which was out of its context, the system returned irrelevant context but the LLM correctly answered from parametric knowledge rather than hallucinating a connection to the retrieved fragments.

Methodology:

Technical Approach

The pipeline is built as a LangGraph [StateGraph](#) with the following node sequence:

Question -> Rewriter -> Retriever -> ReRanker -> Reader -> [Research | Marketing] -> Merge

Key technical choices:

- **Embedding model:** [multi-qa-mpnet-base-dot-v1](#) (sentence-transformers). Selected for its strong performance on question-answering retrieval tasks; purpose-built for asymmetric semantic search where queries and passages have different structures.
- **Vector store:** Qdrant in-memory with cosine similarity. Sufficient for the ~3,900 chunk corpus; avoids infrastructure overhead for this POC.
- **Rewriter:** Rewrites the initial user query using the LLM to improve retrieval by expanding context and standardizing terminology.
- **ReRanker:** Uses a cross-encoder ([BAAI/bge-reranker-v2-m3](#)) to re-evaluate and rerank the initial top-*k* retrieved documents to a more precise top-*n* list based on deep semantic relevance.
- **LLM:** Cohere [command-a-03-2025](#) via API. Selected after the finalist comparison showed it outperformed Mistral-7B-Instruct-v0.3 (4-bit quantized, local) on all metrics while also being 3.5x faster.
- **Rewriter and ReRanker Usage:** Although included in the full pipeline architecture, the experimental evals showed these components degraded performance on this particular corpus. The baseline config without them achieved the highest BERT-F1 and ROUGE-L in the 11-config comparison.

Dual persona accommodation: Both personas share the same retrieval results. Differentiation occurs at the prompt level:

- The *engineering prompt* instructs the model to provide "concrete details and correct terminology" for an expert audience, encouraging inclusion of metrics, algorithms, and architecture details.
- The *marketing prompt* instructs the model to "lead with a single clear value proposition sentence," avoid jargon, and limit output to 4 sentences.

Both prompts share a system prompt requiring all claims to be grounded in context and uncertainty to be stated explicitly.

Testing and Evaluation

I evaluated pipeline performance using four metrics computed against gold reference answers:

- **BERTScore** (F1): Measures semantic similarity using contextual embeddings from [roberta-large](#). This is the primary metric because it is robust to paraphrasing and style shifts between personas.
- **ROUGE** (rouge1, rouge2, rougeL): Measures n-gram overlap. Useful as a secondary signal for lexical grounding, but systematically lower for the marketing persona due to intentional vocabulary divergence.
- **Faithfulness** (BERTScore of answer vs. extracted facts): Measures how well the generated answer is grounded in the retrieved context. My main use case here was to detect hallucination.
- **Length ratio**: Generated answer length divided by gold answer length. Values near 1.0 were used to indicate an appropriate level of verbosity.

My testing strategy proceeded in three phases:

1. **Experimental sweep** (4-question subset, 11 configs): Tested several combinations of rewriter/reranker, chunk sizes (128, 512, 1000, 1500), embedding models (mpnet, miniLM, distilroberta), prompt variants, and quantization modes. Identified “baseline config” as the winner. This used many of the pre-filled parameters given in the notebook.
2. **Finalist comparison** (75-question full validation, 3 configs): Tested the winning baseline, a concise-prompt variant, and a Cohere API variant. Cohere baseline won on all primary metrics.
3. **Test question analysis** (3 questions): Detailed examination of responses, context, sources, and per-question metrics to validate qualitative behavior.

Results and Findings

Proof of Concept Functionality

The finalist comparison on the full 75-question validation set produced:

Config	Persona	BERT-F1	ROUGE-L	Faithfulness	Len Ratio	Time
baseline	Research	0.858	0.214	0.864	1.36	2842s
baseline	Marketing	0.869	0.218	0.865	2.57	
baseline_concise	Research	0.860	0.248	0.869	1.19	2378s
baseline_concise	Marketing	0.868	0.238	0.854	1.93	
cohere_baseline	Research	0.876	0.254	0.894	1.24	816s
cohere_baseline	Marketing	0.881	0.225	0.879	2.08	

The Cohere baseline achieves the best scores on both personas for BERT-F1 and faithfulness, with competitive ROUGE-L. Its research length ratio (1.24) is close to the ideal, and its inference time is 3.5x faster than the Mistral-7B configs.

Lessons Learned

- **Model size matters more than retrieval complexity for this task.** The Cohere Command A model (likely much larger than 7B parameters) produced better answers from the same retrieved context, suggesting that generator quality is the primary bottleneck, not retrieval sophistication.
- **Small chunks trade detail for diversity.** The 128-token chunk size retrieves more diverse snippets across the corpus, which helps aggregate metrics. But individual chunks are too small to contain complete explanations, which hurts on questions requiring specific details.
- **Rate limiting is manageable with engineering.** The Cohere trial key's 20 req/min limit was handled with a sliding-window rate limiter and exponential backoff retry. The entire 75-question evaluation completed in ~14 minutes of wall time, acceptable for a POC.
- **Prompt engineering has diminishing returns.** The concise-prompt variant and the LLM-engineered prompts (from a separate Qwen3 prompt-optimization notebook) both failed to outperform the baseline prompts. Simple, clear instructions with explicit audience framing were more effective than elaborate prompt engineering.

Challenges and Limitations

- **Chunk size of 128 is too aggressive for detail-oriented questions.** Question 0 demonstrated this clearly: the gold answer references "DPR" (Dense Passage Retriever), but the 128-token chunks truncated the source paragraph before that term appeared. The model appropriately hedged rather than hallucinating, but a 500-1000 token chunk would have captured the answer.
- **Out-of-domain queries return irrelevant context.** Question 225 (Marvel actors) exposed that the vector retriever returns the "least bad" matches even when nothing in the corpus is relevant. The Cohere model handled this gracefully, but a less capable model might attempt to ground its answer in the irrelevant context.
- **API dependency introduces operational risk.** The winning config relies on the Cohere API, which is subject to rate limits, latency variability, and potential deprecation. A fallback to local Mistral-7B inference should be maintained.
- **Evaluation metrics don't detect factual errors directly.** BERTScore and ROUGE measure similarity to references, not correctness. A confident but wrong answer could score well if it uses the right vocabulary.

Next Steps

1. **Tune chunk size to 500-750 tokens** with appropriate overlap (100-150) to balance context completeness against retrieval diversity. This is the single highest-impact change.

2. **Add a confidence/relevance gate** that detects when retrieved context is irrelevant (low similarity scores) and routes to a "I don't have enough information" response rather than relying on parametric knowledge.
3. **Implement LLM-as-judge evaluation** to complement BERTScore and ROUGE with factual correctness checks.
4. **Test with a persistent vector store** (Qdrant Cloud or Pinecone) and measure retrieval latency at scale to validate production readiness.
5. **Explore LoRA fine-tuning** on the Mistral-7B model using the 75 gold Q/A pairs per persona, which could close the gap with Cohere while eliminating API dependency.

Summary & Recommendations

The RAG proof of concept successfully demonstrates dual-persona question answering with strong performance on the selected metrics: BERT-F1 above 0.87 and faithfulness above 0.87 for both audiences. The system correctly differentiates between engineering and marketing outputs through prompt design while maintaining grounding in retrieved context.

Recommendation: Proceed to a production pilot using the Cohere API backend with the following conditions:

- **Increase chunk size** from 128 to 500-750 tokens before deployment. The current chunk size produces good aggregate metrics but fails on detail-oriented questions.
- **Deploy as a pay-per-use API** through Cohere. At estimated volumes (50-200 queries/hour), API costs will be under \$100/month versus \$1,500+/month for reserved GPU instances.
- **Maintain Mistral-7B as a fallback** for data-sensitivity scenarios where content cannot be sent to external APIs.
- **Add monitoring** for faithfulness scores in production to detect degradation as the document corpus evolves.

The primary **risk** is API dependency while the primary **value** is the 3.5x speed improvement and superior answer quality compared to local inference. The system is architecturally sound and the retrieval/generation split allows either component to be upgraded independently as better models become available.

References

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- Mistral AI (2024). Mistral-7B-Instruct-v0.3. <https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3>

Appendix

Configuration Eval Results (4-Question Subset)

Config	Persona	BERT-F1	ROUGE-L	Faith	Len-R	Time
optimized	Research	0.831	0.152	0.839	2.13	267.6s
optimized	Marketing	0.869	0.232	0.843	2.90	
baseline	Research	0.866	0.297	0.878	1.13	131.6s
baseline	Marketing	0.862	0.227	0.874	3.66	
no_rewriter	Research	0.831	0.189	0.853	1.68	185.0s
no_rewriter	Marketing	0.874	0.242	0.836	2.59	
no_reranker	Research	0.857	0.247	0.841	2.10	233.4s
no_reranker	Marketing	0.862	0.203	0.828	2.94	
chunk_512	Research	0.849	0.183	0.833	0.69	199.4s
chunk_512	Marketing	0.847	0.191	0.874	2.96	
chunk_1500	Research	0.845	0.234	0.848	1.63	265.1s
chunk_1500	Marketing	0.862	0.259	0.840	2.96	
concise_prompt	Research	0.867	0.290	0.844	1.18	184.1s
concise_prompt	Marketing	0.864	0.227	0.830	2.23	
distil_roberta	Research	0.847	0.212	0.868	1.30	214.2s
distil_roberta	Marketing	0.860	0.192	0.844	2.84	
mini_lm	Research	0.832	0.155	0.852	1.77	237.9s
mini_lm	Marketing	0.865	0.227	0.839	2.62	
engineered_prompts	Research	0.852	0.282	0.883	1.89	334.2s
engineered_prompts	Marketing	0.839	0.174	0.881	6.06	
8bit_quant	Research	0.839	0.150	0.828	1.19	512.3s
8bit_quant	Marketing	0.880	0.262	0.836	2.50	

Error Discussion

Question 0 (missed detail): The gold answer identifies "DPR" (Dense Passage Retriever) as the specific non-parametric memory component. Our system's 128-token chunks truncated the relevant paragraph from Lewis et al. (2020) before this detail appeared. The model demonstrated good faithfulness by hedging ("the context does not explicitly specify...") rather than guessing, but a larger chunk size would have captured the answer.

Question 225 (out-of-domain): The system returned completely irrelevant context (author names, fragment phrases from ML papers) for a pop-culture question. The Cohere model correctly ignored the context and answered from parametric knowledge ("Josh Brolin"). This is desirable behavior for a production system, but it means the answer is unverifiable against the corpus -- a confidence score or relevance gate would help users distinguish RAG-grounded answers from parametric fallbacks.

Pipeline Errors: I also ran into several errors from hg uploads timing out and not saving the final versions of these cells, but they are included in the notebook and report as they were written at the time of the final evaluation runs.